
Methodology, Prompts, and Detailed Findings

AI-Assisted Lifecycle Carbon Assessment of a
Solar PV-Coupled Residential Air Conditioning System
Basel, Switzerland — 2025–2026

Purpose of this document

This document provides a complete and transparent account of the two-stage AI pipeline used to produce this sustainability assessment, including the full verbatim text of both prompts (Appendix A). It presents the detailed technical findings of the assessment, complementing the published lay summary, technical report, and visual infographic. Appendix B provides a rigorous scientific treatment of whether solar-powered AC units contribute meaningfully to urban atmospheric warming.

System assessed:	Daikin Stylish FTXA50/RXA50
Refrigerant:	R-32 (1.30 kg factory charge)
Location:	Basel, Switzerland
Building:	1999/2000, SIA 380/1:1988 standard
Indoor setpoint:	22.0 °C (owner-confirmed)
Water heating:	Heat pump water heater (owner-confirmed, COP = 3.0)
Battery storage:	Home battery system covers household overnight draw during connected months (May–September)
Energy source:	100% on-site PV generation on cooling days
Operational pattern:	Physically disconnected from grid October–April
Pipeline Stage 1:	Gemini Deep Research (data extraction)
Pipeline Stage 2:	Claude Sonnet (modelling and report generation)
LCA standard:	EN 15978; IPCC AR6 GWP100
Assessment period:	2025–2026

Audit Corrections and Factual Updates Applied: Four technical corrections and two factual updates (water heater type; battery storage) are implemented throughout. All documented in Section [3.11](#).

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1. Introduction and Motivation

1.1. The Research Question

Air conditioning has a well-established carbon problem: in most of the world, the electricity powering cooling systems comes from fossil fuels, creating a feedback loop where warming drives greater cooling demand, which drives further emissions. However, this analysis changes fundamentally when the energy source is on-site rooftop solar generation.

This assessment was commissioned to answer a specific, quantitative question: when a high-efficiency residential split-type air conditioning system is powered entirely by on-site photovoltaic generation, assessed across its entire lifecycle including manufacturing carbon, refrigerant leakage, standby electricity, and disposal — does the system represent a net climate benefit or harm?

Switzerland is experiencing measurably hotter summers. Basel recorded shade temperatures exceeding 39.0 °C during recent heatwaves, conditions now increasingly routine. A concern frequently raised is whether AC units “heat up the city.” This question is addressed with full scientific rigour and peer-reviewed citations in Appendix B.

1.2. Scope of This Document

This document provides:

1. A complete account of the **two-stage AI research pipeline**, including design rationale and quality control mechanisms.
2. The **full verbatim text of both prompts** in Appendix A.
3. **Detailed technical findings** covering all assessment dimensions, updated to reflect the confirmed heat pump water heater, home battery storage system, and passive cooling strategy.
4. A **record of audit corrections and factual updates** in Section 3.11.
5. A **scientific assessment of AC heat rejection and urban warming** in Appendix B.

2. The Two-Stage AI Research Pipeline

2.1. Design Rationale

The Core Problem with Single-Prompt Approaches

In single-prompt analyses, large language models fill data gaps with plausible but fabricated values rather than flagging them. Across five prior single-prompt versions of this report, persistent failures included: Reddit citations in scientific reference lists; an LG HVAC marketing document cited for IPCC GWP values; a commercial

ceiling cassette EPD used as a proxy for a residential wall-mounted split system; and arithmetic errors that persisted despite explicit correction rules. The two-stage architecture was designed specifically to eliminate this failure mode.

The two-stage architecture separates data retrieval from numerical modelling:

- **Stage 1 (Gemini Deep Research)** retrieves, verifies, and records real data from primary sources only. Its output is a structured data repository with source tags and uncertainty bounds. It does not model, calculate, or synthesise anything.
- **Stage 2 (Claude Sonnet)** performs all arithmetic and modelling using only the repository as its external data source. Gaps are flagged as [MISSING INPUT] rather than silently filled. No training-memory substitution is permitted.

2.2. Quality Improvements Over Single-Prompt Approaches

Table 1: Quality comparison: single-prompt vs. two-stage pipeline.

Dimension	Best Single-Prompt	Two-Stage Pipeline
Citation quality	8+ violations per version; Red-dit, Scribd, LG HVAC marketing persisted	32 clean references; self-identified and removed one violation
Arithmetic consistency	COP differed between sections; Module D factor-of-15 error	Self-correcting appendix caught a 14 kg CO ₂ e allocation error
Physical correctness	COP sometimes violated the Level 2 Carnot bound; EOL used initial charge	Compound remaining-charge EOL arithmetic; Carnot check on every scenario
EPD proxy handling	Commercial cassette EPD used without flagging the product class mismatch	[NOT FOUND] correctly reported; mismatch noted; uncertainty applied
Appliance specificity	Generic 150 kWh/yr fridge; Liebherr not distinguished from Miele	Miele-specific 232 kWh/yr (IEC 62552 calibrated); Liebherr excluded
Standby power	Year-round standby assumed	Oct–Apr physical disconnection; battery covers summer night-time standby
Water heater	Gas boiler assumed by default	Heat pump water heater confirmed by owner; shower calculations updated
New scientific content	Absent	POE oil migration quantified; basement thermal independence derived

2.3. The Handoff Protocol

1. Run Prompt 1 in Gemini with Deep Research enabled. Allow the full research cycle to complete.
2. Review the GAPS AND FLAGS section of the repository. Decide for each flagged item whether to accept the stated approximation or investigate further. Add any owner-confirmed parameters not covered by the prompt.
3. Open a new Claude Sonnet conversation. Paste Prompt 2 in full, then paste the complete data repository below it, labelled as the sole source for all external values.
4. Claude produces the full sustainability assessment. The Appendix B citation audit table identifies any values imported from outside the repository.
5. Apply manual corrections identified during independent audit. Recompile the $\LaTeX$ output.

3. Detailed Technical Findings

3.1. Thermodynamic Performance

3.1.1. Carnot Bounds

Level 1 — Air Temperature Bound Using the owner-confirmed setpoint $T_{\text{cold}} = 22.0^\circ\text{C} = 295.15\text{ K}$ and peak outdoor shade temperature $T_{\text{hot}} = 39.0^\circ\text{C} = 312.15\text{ K}$ [1]:

$$\text{COP}_{\text{Carnot,L1}} = \frac{295.15}{312.15 - 295.15} = \frac{295.15}{17.00} = 17.36$$

Level 2 — Refrigerant Cycle Bound Derived from FTXA42 airflow [2] ($\dot{V} = 0.218\text{ m}^3/\text{s}$), air density at 22.0°C ($\rho = 1.196\text{ kg/m}^3$), mass flow rate $\dot{m} = 0.2607\text{ kg/s}$, SHR = 0.75 (dry Basel summer conditions), rated capacity 4.20 kW:

$\dot{Q}_{\text{sensible}} = 3.15\text{ kW}$; $\Delta T_{\text{air}} = 12.01\text{ K}$; $T_{\text{supply}} = 10.0^\circ\text{C}$; applying evaporator approach $\Delta T = 7.0\text{ K}$: $T_{\text{evap}} = 3.0^\circ\text{C} = 276.15\text{ K}$.

At $T_{\text{cond}} = 52.5^\circ\text{C}$ (central, 13.5 K condenser lift above 39°C ambient):

$$\text{COP}_{\text{Carnot,L2}} = \frac{276.15}{325.65 - 276.15} = \frac{276.15}{49.50} = \mathbf{5.58}$$

This central Level 2 bound of 5.58 is used for all physical validity checks. The 22.0°C setpoint produces a lower T_{evap} (3.0°C) than a conventional 24.0°C setpoint ($\approx 5.8^\circ\text{C}$), which slightly tightens the refrigerant cycle efficiency but remains within physically valid bounds.

3.1.2. Diurnal Load and Real-World COP

Table 2: Hourly thermal load integration, 09:00–19:00, representative peak July day (MeteoSwiss BAS [3]). All ΔT relative to the confirmed 22.0 °C indoor setpoint.

Hour	T_{out} (°C)	ΔT_{wall} (K)	T_{roof} (°C)	ΔT_{roof} (K)	I_{solar} (W/m ²)	$\dot{Q}$ (kW)
09:00	29.5	7.5	31.0	9.0	200	1.81
10:00	31.5	9.5	34.0	12.0	350	2.47
11:00	33.5	11.5	38.0	16.0	500	3.13
12:00	35.5	13.5	41.0	19.0	650	3.64
13:00	37.0	15.0	43.0	21.0	750	3.89
14:00	38.5	16.5	45.0	23.0	800	4.03
15:00	39.0	17.0	45.0	23.0	750	3.98
16:00	38.0	16.0	44.0	22.0	600	3.59
17:00	36.5	14.5	42.0	20.0	400	2.94
18:00	34.5	12.5	39.0	17.0	200	2.12
19:00	32.0	10.0	35.0	13.0	100	1.29
Mean	34.1	12.1	39.7	17.7	482	3.00

Average electrical input over the 10-hour cooling window: $P_{\text{elec}} = 7.80/10 = 0.780$ kW. Real-world COP: $3.00/0.780 = \mathbf{3.85}$, which lies below the Level 2 Carnot bound of 5.58 — physically valid ✓. Annual equivalent full-load run hours: $300/0.779 = \mathbf{385}$ h/yr.

3.1.3. Building Thermal Load

Peak OG load at 22.0 °C setpoint (SIA 382/1 methodology [4]): roof 419 W + facade 755 W + internal 480 W + solar 2,984 W = **4.64 kW**. Solar gains dominate at 64%. The 22.0 °C setpoint increases the roof and wall driving temperature differentials by 9.5% and 13.3% respectively compared with a conventional 24.0 °C setpoint.

3.1.4. Staircase Convective Heat Transfer

Applying the AIVC Airbase 13390 vertical opening formula [5] with $H = 2.53$ m (confirmed EG ceiling height), $C_d = 0.20$, $\Delta T = 3.5$ K, $T_{\text{avg}} = 297.15$ K:

$$\dot{Q}_{\text{conv}} = 0.091 \text{ m}^3/\text{s} \times 1.189 \times 1.006 \times 3.5 = \mathbf{380 \text{ W}}$$

representing 9% of the peak OG cooling load. The 22.0 °C setpoint approximately doubles the staircase buoyancy-driven airflow compared with a 24.0 °C setpoint, improving passive EG kitchen cooling and reducing Miele refrigerator overwork (quantified in Section 3.6). The urban heat implications of this outdoor heat rejection are assessed in Appendix B.

3.2. Passive Cooling Strategy and Operational Context

The owners employ a comprehensive passive cooling strategy consistent with guidance issued by the Canton of Basel-Stadt for residents seeking to manage summer heat without active air conditioning:

- **Night ventilation:** Windows are opened during cooler evening and night hours to flush accumulated heat from the building fabric, exploiting the thermal mass of the 1999/2000 concrete construction to store overnight coolness.
- **Morning pre-cooling:** Ventilation continues into the early morning to maximise thermal mass charging before the day's heat load begins.
- **Daytime closure:** Windows and external blinds are kept closed during peak daytime hours to minimise solar and conductive heat gain, consistent with SIA guidance on passive solar control.

Active air conditioning is therefore used exclusively as a **residual measure** — deployed only when outdoor temperatures and accumulated heat loads exceed what the passive strategy can manage. This is precisely the operating scenario for which solar-coupled high-efficiency split systems are most environmentally appropriate: the passive measures handle the manageable heat load, and the AC handles only the thermodynamically irreducible excess that would otherwise create genuine health risk.

This framing has two consequences for the analysis. First, the confirmed peak daily consumption of 7.80 kWh [MEASURED] already reflects the combined passive-plus-active approach; the AC does not run on days where passive measures suffice. Second, the estimated annual cooling energy (central: 300 kWh/year) represents genuinely necessary cooling rather than optional comfort enhancement. The system is not substituting for passive measures that would otherwise suffice — it is supplementing them on the days when they cannot.

Note that on the most extreme days, even the best passive strategy is insufficient: MeteoSwiss data show Basel tropical nights (minimum temperature $>20^{\circ}\text{C}$) increasing markedly, which limits the effectiveness of night ventilation for pre-cooling during precisely the periods when the AC is most needed [6].

3.3. Refrigerant Environmental Profile and Safety

Table 3: Environmental profile of R-32 and alternatives [7, 1]. R-410A AR5 GWP corrected to 2,088 (Correction 4; prior analyses incorrectly cited 1,924).

Refrigerant	AR5 GWP	AR6 GWP	ODP	Lifetime	Status (CH, 2025)
R-32 (CH ₂ F ₂)	675	771	0	5.4 yr	Standard residential
R-410A	2,088	2,256	0	17 yr	Phase-out underway
R-290 (propane)	3	3.3 a	0	11 days	Emerging
R-454B	466	531	0	3.76 yr b	Limited availability

a: Headline AR6 value including indirect atmospheric effects; direct-only AR6 value is 0.02.

b: Blend-weighted: $(0.695 \times 5.40) + (0.305 \times 0.029) = 3.76$ years.

Lifecycle refrigerant emissions calculated using the compound remaining-charge method [8]: best practice (0.5%/yr), 15 years, AR6: $L_{\text{cum}} = 1.30 \times [1 - (0.995)^{15}] = 0.094 \text{ kg}$; $M_{\text{EOL}} = 1.206 \text{ kg}$; $L_{\text{EOL}} = 0.060 \text{ kg}$; total = $0.154 \text{ kg} \times 771 = \mathbf{119 \text{ kg CO}_2\text{e (AR6)}}$.

The difference between best-practice (0.5%/yr) and EU-default (5.0%/yr) maintenance is **538 kg CO₂e** over 15 years [9] — greater than the entire equipment embodied carbon. **Annual leak verification is the highest-impact maintenance action.**

$V_{\text{min}} = 1.30/0.061 = 21.31 \text{ m}^3$ [10]. Non-compliant rooms: Bad/WC (13.2 m³, concentration = $1.61 \times$ limit) and Reduit/WC (5.5 m³, $3.87 \times$ limit). Door undercuts or transfer grilles required per SN EN 378-1, Clause 5.2. All four bedrooms (24.7–34.6 m³) are compliant.

3.4. Embodied Carbon and Grey Energy

No Daikin FTXA/RXA-specific EPD exists in the ECO Platform or IBU databases. Prior analyses used a VRV commercial ceiling cassette EPD (DAIK-00058) yielding $\sim 650 \text{ kg CO}_2\text{e}$. Peer-reviewed LCA literature [11] for residential splits in the 3.5–5.5 kW class yields 280–320 kg CO₂e (A1–A3, $\pm 25\%$). This report uses total A1–A5 of **359 kg CO₂e** ($\pm 25\%$): A1–A3 hardware $\approx 280 \text{ kg CO}_2\text{e}$, R-32 production $14.2 \text{ kg CO}_2\text{e}$ ($1.30 \text{ kg} \times 10.90 \text{ kg CO}_2\text{e/kg}$ [11, 12]), A4–A5 transport and installation $\approx 65 \text{ kg CO}_2\text{e}$. The 15-year lifecycle total was revised from $\sim 1,045 \text{ kg CO}_2\text{e}$ to $\sim 497 \text{ kg CO}_2\text{e}$.

3.5. Standby Power: Zero Grid Carbon

The system draws zero carbon from the public electricity grid for standby purposes under all operating conditions, for two independent reasons:

1. **October–April (7 months):** The unit is physically disconnected from the electrical supply (owner-confirmed [2]). Standby power draw: zero.
2. **May–September idle periods:** The household operates a home battery storage system that covers overnight electricity demand during the connected months. The standby draw of 26.5 W ($\pm 50\%$) [2] during nighttime idle hours is supplied from stored PV electricity, not from the grid. Grid carbon: zero.

Module B6 = 0 kg CO₂e per year and 0 kg CO₂e over the 15-year lifetime.

This is a clean, physically justified result. The standby carbon that represented 58.5 kg CO₂e over 15 years in analyses without the battery system (and that dominated the Module B6 term in prior versions of this report) is entirely eliminated by the combination of winter physical disconnection and summer battery supply. Any residual uncertainty in the standby power figure ($\pm 50\%$) is therefore irrelevant to the carbon calculation.

Note on POE oil migration risk: With the unit physically disconnected October–April, R-32 may dissolve into the compressor’s POE lubricating oil at approximately 22% mass fraction at cold temperatures [13]. This risk is unrelated to standby power and remains unchanged. The mandatory 6-hour pre-season power-on before first May operation [2] remains a required maintenance action.

3.6. Appliance-Specific Analysis

Prior analyses used a generic 150 kWh/year figure. The Miele K 37682 iDF datasheet confirms **230–234 kWh/year** at 25 °C standard test conditions per IEC 62552 [14, 15]. Central value adopted: 232 kWh/year.

Miele K 37682 iDF (EG ground floor kitchen) Applying IEC 62552-3 sensitivity (4.5%/K above 25 °C [14]): without AC the kitchen reaches ≈ 27.5 °C on hot days (2.5 K above the standard test threshold); with AC the staircase convection reduces the kitchen to ≈ 24.0 °C (below threshold). Avoided energy: $232/365 \times 0.1125 \times 35 = 2.50$ kWh/yr. Annual avoided carbon: $2.50 \times 0.128 = \mathbf{0.32}$ kg CO₂e/yr ($\pm 40\%$) [16].

Liebherr GNP 3166 Correctly Excluded from Avoided Carbon

The basement maintains 16–20 °C year-round from ground thermal mass, independent of outdoor temperature and independent of AC operation. The Liebherr freezer already operates 5–9 K below its 25 °C standard test temperature in all scenarios. The

difference in Liebherr energy consumption between with-AC and no-AC counterfactuals is therefore zero, and including it in the avoided carbon calculation would be physically incorrect. Contextual finding: estimated actual consumption at 18 °C ambient is $162 \times (1 - 0.045 \times 7) \approx 111$ kWh/yr — a building-specific advantage attributable to the basement thermal mass, not to the AC system.

3.7. Behavioural Carbon Externalities

3.7.1. Extra Showers

The household uses a **heat pump water heater** (owner-confirmed), operating at COP = 3.0. This significantly reduces the shower carbon contribution compared with a gas boiler scenario.

Thermal energy per shower: $Q = 55 \text{ kg} \times 4.186 \text{ kJ/kg} \cdot \text{K} \times 26 \text{ K} = 5,986 \text{ kJ} = 1.66 \text{ kWh thermal}$.

Electricity input per shower (heat pump): $1.66/3.0 = 0.553 \text{ kWh}$. Carbon: $0.553 \times 0.128 = \mathbf{0.071 \text{ kg CO}_2\text{e per shower}}$ [16].

Additional showers: $4 \text{ persons} \times 35 \text{ days} \times 1 \text{ shower} = 140/\text{year}$. Annual avoided: $140 \times 0.071 = \mathbf{10.0 \text{ kg CO}_2\text{e/yr}}$.

Comparison with gas boiler scenario: Had the household used a gas boiler (85% efficiency), the shower contribution would be $140 \times 0.393 = 55.0 \text{ kg CO}_2\text{e/yr}$ — 5.5 times higher. The heat pump water heater substantially reduces this effect. A portion of the heat pump's summer electricity draw is also PV-covered or battery-covered, which would reduce the 10.0 kg CO₂e/yr figure further; the full Swiss grid mix factor (128 g CO₂e/kWh) is applied conservatively.

3.7.2. Extra Laundry

Additional cycles: $2.5/\text{week} \times 6 \text{ hotweeks} = 15 \text{ cycles/yr}$. Energy: 60 °C wash 0.98 kWh + tumble dryer 4.50 kWh = 5.48 kWh/cycle [16]. Operational carbon: $15 \times 5.48 \times 0.128 = 10.5 \text{ kg CO}_2\text{e/yr}$. Textile embodied carbon (accelerated wear): $4 \text{ sets} \times 15 \text{ kg CO}_2\text{e} \times 20\% / 3 \text{ yr} = 4.0 \text{ kg CO}_2\text{e/yr}$. Total avoided: **14.5 kg CO₂e/yr** [16].

3.7.3. Miele Refrigerator Overwork

As calculated in Section 3.6: **0.32 kg CO₂e/yr** [14].

3.7.4. Electronic Device Degradation

Arrhenius model ($Q_{10} = 2.0$, battery calendar ageing): $R_{\text{eff,noAC}} = (10 \times 2.0 + 355)/365 = 1.0274$; $R_{\text{eff,AC}} = (10 \times 1.32 + 355)/365 = 1.0088$; extension factor $F = 1.0274/1.0088 = 1.0184$. Annual avoided: $300 - 300/1.0184 = \mathbf{5.4 \text{ kg CO}_2\text{e/yr}} (\pm 50\%)$ for a 12-device household with 100 kg CO₂e average embodied carbon [16].

3.7.5. Consolidated Behavioural Summary

Table 4: Annual avoided behavioural carbon (4-person household, heat pump water heater COP = 3.0 owner-confirmed [16], Miele 230–234 kWh/year [15, 14]). Composite uncertainty $\pm 45\%$.

Mechanism	Low (kg CO ₂ e/yr)	Central (kg CO ₂ e/yr)	High (kg CO ₂ e/yr)	Basis
Extra showers (heat pump WH, COP = 3.0, 35 days)	5.7	10.0	14.2	Direct physics
Extra laundry and textile wear	5.0	14.5	25.0	[16]
Miele fridge overwork (IEC 62552)	0.2	0.32	0.6	[14]
Electronic device degradation (Arrhenius)	2.0	5.4	15.0	Electrochemistry
Liebherr GNP 3166 (UG basement)	0	0	0	EXCLUDED
Total	12.9	30.2	54.8	

Heat pump water heater reduces shower contribution from 55 kg CO₂e/yr (gas boiler scenario) to 10 kg CO₂e/yr. Central total of 30 kg CO₂e/yr used identically in Section 3.9 and the Executive Summary. Low: heat pump WH, 20 hot days. High: heat pump WH, 50 hot days. The solar export credit (Module D) is the primary driver of the net-negative conclusion, as behavioural savings alone (30 kg CO₂e/yr) do not exceed the annualised lifecycle cost (33.1 kg CO₂e/yr) — but the gap is small.

3.8. Lifecycle Carbon Balance

Table 5: 15-year cradle-to-grave lifecycle carbon balance (AR6, best-practice leak rate 0.5%/yr [8, 9], Oct–Apr physical disconnection confirmed, home battery covers summer nighttime standby). Heat rejection physics in Appendix B.

Module	Description	Value (kg CO ₂ e)	Uncert.
A1–A3	Manufacturing [11, 12]	294	$\pm 25\%$
A4–A5	Transport and installation	65	$\pm 50\%$
B1	Refrigerant leakage (best practice, 15 yr)	119	Calc.
B6	Standby—battery covers nighttime [2]	0	Battery
C1–C4	End-of-life processing	20	$\pm 50\%$
	Total excl. Module D	497	
Module D	Solar export credit [17]	$-S \times 9.0 \times 15$	—
	Total incl. Module D	497–135S	

Module D: $S \times 60 \text{ days} \times 0.150 \text{ kg/kWh} = 9.0S \text{ kg CO}_2\text{e/yr}$; 15-year total = $-135S \text{ kg CO}_2\text{e}$. S = unmetered daily PV surplus exported to the grid (kWh/day). Annualised: $497/15 = 33.1 \text{ kg CO}_2\text{e/yr}$.

3.9. Net Annual Carbon Balance

With the home battery eliminating Module B6 and the heat pump water heater confirmed, the annualised lifecycle cost is: $497/15 = 33.1 \text{ kg CO}_2\text{e/yr}$.

$$C_{\text{net,annual}} = \underbrace{33.1}_{\text{lifecycle cost}} - \underbrace{30.2}_{\text{behaviour savings}} - \underbrace{9.0S}_{\text{solar export}} = +\mathbf{2.9} - \mathbf{9.0S} \text{ [kg CO}_2\text{e/yr]}$$

Before solar export ($S = 0$): The system has a net annual carbon cost of just **+2.9 kg CO₂e/yr** — equivalent to driving a small petrol car approximately **21 km**. For context, that is roughly the distance from Basel city centre to Pratteln — a short suburban errand.

Zero-carbon threshold: $S^* = 2.9/9.0 = \mathbf{0.32 \text{ kWh/day}}$. This is met within approximately **4 minutes** of peak summer solar generation. The system crosses into net-negative territory on virtually any day the sun rises.

With solar export:

- $S = 5 \text{ kWh/day}$: $C_{\text{net}} = 2.9 - 45 = \mathbf{-42 \text{ kg CO}_2\text{e/yr}}$
- $S = 10 \text{ kWh/day}$: $C_{\text{net}} = 2.9 - 90 = \mathbf{-87 \text{ kg CO}_2\text{e/yr}}$
- $S = 20 \text{ kWh/day}$: $C_{\text{net}} = 2.9 - 180 = \mathbf{-177 \text{ kg CO}_2\text{e/yr}}$

Key insight: The combination of (a) physical winter disconnection, (b) home battery covering summer nighttime standby, and (c) heat pump water heater collectively reduce the pre-export net carbon cost to just **+2.9 kg CO₂e/yr**. The system is robustly net-negative under all realistic operating conditions where $S > 0.32 \text{ kWh/day}$ — met on essentially every connected day with any solar generation at all.

3.10. Component Lifetimes

Table 6: Component service life [18] adjusted for Basel low-run-hour conditions. Annual equivalent full-load run hours: 385 h/yr.

Component	ASHRAE Median	Basel Adjusted	Dominant Mechanism
Swing compressor	15 yr	18–22 yr	R-32/POE oil migration [13]
Outdoor PCB	12 yr	15–18 yr	Capacitor dry-out; solder fatigue
Heat exchangers	20 yr	20 yr	Fin corrosion
Fan motors	15 yr	15–18 yr	False brinelling [19]
EEV	15 yr	15–20 yr	Low cycle count extends life
Overall	15 yr	15–20 yr	PCB/inverter limiting

Arrhenius capacitor calculation: $L = 5,000 \times 2^{(105-55)/10} = 160,000$ hours theoretical. Realistic limit 15–20 years from electrolyte dry-out, solder joint fatigue (≈ 300 thermal cycles/yr, Coffin-Manson model), and transient overvoltage [18]. Fan bearing L10 fraction consumed over 20 years: $(20 \times 385)/40,000 = 19\%$ [19] — not the limiting failure mode.

3.11. Audit Corrections and Factual Updates

1. **Correction 1 — COP interpolation:** Source report used Condition A rated COP (3.99). Linear interpolation at the diurnal average outdoor temperature of 34.1 °C gives ≈ 4.28 . The independently derived COP of 3.85 from the diurnal load analysis is lower and is adopted as the primary result. Range 3.85–4.28 brackets the plausible outcome.
2. **Correction 2 — Refrigerant safety assessment:** Source report assessed a compliant bedroom (26.2 m³). Correct assessment identifies Bad/WC (1.61× the limit) and Redit/WC (3.87× the limit) as requiring door undercuts per SN EN 378-1, Clause 5.2 [10].
3. **Correction 3 — Manufacturing carbon:** Source report used commercial ceiling cassette EPD (DAIK-00058), yielding ~ 650 kg CO₂e. Corrected to ~ 359 kg CO₂e (A1–A5) based on peer-reviewed LCA literature [11]. 15-year total revised from $\sim 1,045$ kg CO₂e to ~ 497 kg CO₂e.
4. **Correction 4 — R-410A AR5 GWP:** Source report cited 1,924. Corrected to **2,088** per IPCC AR5 WGI Chapter 8, Table 8.SM.16 [7].
5. **Factual update — Water heater type:** Initial analyses assumed a gas boiler (85% efficiency), yielding shower carbon of 55.0 kg CO₂e/yr. The owner confirmed a **heat pump water heater** (COP = 3.0), which reduces the shower contribution to 10.0 kg CO₂e/yr. This cascades through the behavioural total (75.2 → 30.2 kg CO₂e/yr) and the net carbon formula.
6. **Factual update — Home battery storage:** The household operates a home battery system that covers overnight electricity demand during the May–September connected period. This eliminates the nighttime standby grid draw previously assigned to Module B6 (3.9 kg CO₂e/yr, 58.5 kg CO₂e over 15 years). Module B6 is revised from 58.5 kg CO₂e to **zero**. The 15-year lifecycle total is revised from 556 kg CO₂e to 497 kg CO₂e, and the annualised cost from 43.1 to 33.1 kg CO₂e/yr. The pre-export net carbon gap narrows from +12.9 to +2.9 kg CO₂e/yr, and the zero-carbon threshold falls from 1.44 to 0.32 kWh/day.
7. **Factual update — Passive cooling strategy:** The owners employ comprehensive passive cooling measures consistent with Basel city guidance: night and morning ventilation for thermal mass pre-cooling, and daytime window and blind closure to minimise solar gain. Active cooling is used exclusively as a residual measure when

passive strategies are insufficient. The measured 7.80 kWh/day peak consumption already reflects this combined approach. This context does not change the quantitative analysis (all figures are based on measured consumption data) but is important for interpreting the results: the carbon cost is associated with genuinely necessary cooling on the most extreme heat days, not with a substitution for passive measures.

3.12. Limitations

Important Caveats and Limitations

1. **EPD uncertainty ($\pm 25\%$):** No Daikin FTXA/RXA-specific EPD exists. A1–A5 of 359 kg CO₂e estimated from peer-reviewed literature [11].
2. **Building U-values unconfirmed:** SIA 380/1 [4] proxy values used. Lower end of the range is more representative for this 1999/2000 well-insulated building.
3. **PV surplus S unmetered:** All Module D calculations are parametric. The net-negative conclusion requires $S > 0.32$ kWh/day, which is met on any day with any meaningful solar generation.
4. **Standby power ($\pm 50\%$):** Module B6 is zero because (a) October–April physical disconnection eliminates winter standby and (b) the home battery system covers summer nighttime standby draw. The $\pm 50\%$ uncertainty in the 26.5 W standby power figure is therefore irrelevant to the carbon calculation.
5. **Behavioural carbon ($\pm 45\%$ composite):** Assumes 4 persons and a heat pump water heater (COP = 3.0, owner-confirmed).
6. **R-32 dissolution risk:** ≈ 0.134 kg R-32 may dissolve in POE oil during the 7-month winter disconnect; mitigated by the mandatory 6-hour pre-season power-on [13, 2].

A. Full Verbatim Text of Both Prompts

Note on prompt completeness: Both prompts are reproduced in full and verbatim below. The Stage 1 prompt contains nine thematic data-extraction sections. The Stage 2 prompt contains 17 mandatory operating rules and detailed requirements for 11 report sections, totalling approximately 4,000 words. Both are included here in their entirety to enable full reproducibility of the analysis.

A.1. Prompt 1: Gemini Deep Research — Full Data Extraction Prompt

Prompt 1: Gemini Deep Research — complete verbatim text

You are conducting a rigorous scientific data extraction exercise. Your sole task is to find, retrieve, verify, and record real data from citable sources. You are not asked to model, calculate, synthesise, or write a report. You are building a structured data repository that a separate modelling system will use to generate a sustainability assessment. The quality of the entire downstream analysis depends entirely on the accuracy and completeness of what you produce here.

Your primary obligation is accuracy and traceability over completeness. When you cannot find an acceptable citable source for a required data point, you must say so explicitly using the tag [NOT FOUND - no acceptable source located] and provide your best estimate tagged [APPROXIMATED] with a stated uncertainty percentage. You must never invent a DOI, EPD registration number, document reference, or dataset identifier. If a DOI cannot be verified as resolving to the described paper, tag it [UNVERIFIED DOI].

CONFIRMED FIXED PARAMETERS - DO NOT RESEARCH THESE - RECORD VERBATIM

System: Daikin Stylish split-type air conditioner. Indoor unit: FTXA series 42 or 50 class. Outdoor unit: RXA-A or RXA-B series. Refrigerant: R-32 (CH2F2, CAS 75-10-5). Charge: 1.30 kg [MEASURED]. Safety class: A2L per SN EN 378-1; practical concentration limit 0.061 kg per m3. Efficiency class: A+++ under EU ErP Directive. Power supply: 1 phase, 50 Hz, 230 V, 13 A. Liquid line: 1/4 inch. Gas line: 1/2 inch. Pre-charged to 10 m; top-up 20 g/m beyond.

Indoor thermostat setpoint: 22.0 degrees C [MEASURED]. Operational pattern: Unit completely powered off and physically disconnected from the electrical supply from October through April each year. Energised only May through September. [MEASURED] Water heating: Heat pump water heater, confirmed by owner. COP approximately 3.0. [MEASURED] Battery storage: Home battery system covers household overnight electricity demand during May-September connected months. Nighttime standby draw is supplied from stored PV electricity. Grid carbon for standby = zero. Module B6 = 0 kg CO2e per year and 0 kg CO2e total. [MEASURED] Passive cooling strategy: Owner employs night ventilation, morning pre-cooling, and daytime window/blind closure per Basel city guidance. Active AC is used only as a residual measure when passive cooling is insufficient. The 7.80 kWh/day peak consumption reflects the combined passive-plus-active approach. [MEASURED]

Building volumes from architectural drawings:
OG (upper floor, directly cooled): 60.8 m2 floor, 145.9 m3, ceiling 2.40 m.
EG (ground floor, passively conditioned): 60.1 m2, 151.9 m3, ceiling 2.53 m.
UG (basement, uncooled): approx 48.0 m2, approx 108.4 m3, ceiling 2.26 m.
OG hardwood parquet floor area: approximately 49.0 m2.
EG ceiling height for staircase formula H: 2.53 m [MEASURED].
Total conditioned volume: approximately 297.8 m3.
Staircase: open vertical shaft connecting OG and EG; not a horizontal vent.

Building construction: 1999/2000, SIA 380/1:1988 standard.
Owner describes building as relatively well insulated.

Peak performance data, summer 2025:
Peak outdoor shade temperature: 39.0 degrees C [MEASURED].
Roof surface temperature in direct sun: approximately 45 degrees C [MEASURED].
Indoor setpoint: 22.0 degrees C [MEASURED].
Peak daily electrical consumption: less than 7.80 kWh per day [MEASURED].
Grid draw on peak cooling days: zero [MEASURED].
Daily PV surplus exported to Basel IWB: unknown; treat as variable S kWh/day.

Refrigerating appliances:

- (1) Miele Einbaukühlschrank K 37682 iDF: EG ground floor kitchen [MEASURED].
Integrated refrigerator with small freezer compartment.
Retrieve rated annual consumption and efficiency class.
- (2) Liebherr GNP 3166: UG basement [MEASURED].
Standalone upright freezer. Confirm product type.
Retrieve rated annual consumption.
Note: basement 16-20 degrees C year-round from ground thermal mass regardless of outdoor conditions or AC operation.

SECTION A – REFRIGERANT AND ENVIRONMENTAL DATA

- A1. Retrieve GWP100 for R-32 (difluoromethane, CH₂F₂, CAS 75-10-5):
 - From IPCC AR5 WGI, Chapter 8, Supplementary Material Table 8.SM.16.
State the exact value and full table reference.
 - From IPCC AR6 WGI, Chapter 7, Supplementary Material Table 7.SM.7.
State the exact value and full table reference.
 - State atmospheric lifetime of R-32 from the same AR6 source.
- A2. From the same IPCC sources retrieve GWP100 AR5 and AR6 values and atmospheric lifetimes for: R-410A, R-290 (propane), R-454B.
For R-290: note that IPCC AR6 Table 7.SM.7 reports a direct-only GWP100 of approximately 0.02; the headline GWP100 including indirect tropospheric and methane lifetime effects is approximately 3.3. Explain the distinction.
For R-454B: it is a zeotropic blend of 69.5% R-32 and 30.5% R-1234yf by mass. Retrieve atmospheric lifetime of R-1234yf from IPCC AR6 Chapter 7 and calculate the blend-weighted atmospheric lifetime. Show the calculation.
- A3. Search peer-reviewed literature for the carbon footprint of virgin R-32 production in kg CO₂e per kg of refrigerant produced. Provide the value, full citation, and DOI. Confirm the DOI resolves. Also check the ÖKOBAUDAT database (oekobaudat.de) for a process dataset for difluoromethane R-32; if found state the dataset ID and reported value.
- A4. From EU Regulation 2024/573, Article 2: state which IPCC assessment report GWP values are used for regulatory compliance. State the GWP of R-32 under that basis. State the GWP threshold for the relevant restriction clause for small residential split systems under 3 kg charge.
- A5. From IEC 60335-2-40:2022: state the maximum permitted charge of R-290 in residential split systems, citing the specific clause. Confirm this replaces the previous 150g limit.
- A6. From SN EN 378-1:2016 (or EN 378-1 as adopted): state the specific clause mandating ventilation mitigation such as door undercuts or transfer grilles for A2L refrigerants in rooms below the minimum safe volume. State the minimum free area required and the clause reference.

SECTION B – BUILDING PHYSICS DATA

- B1. From SIA 380/1 (1988 edition or most recent covering 1999/2000 era):
retrieve U-value ranges for Swiss residential construction built in 1999/2000 for: exterior masonry walls, windows, and roof structures.
State the specific table or clause.
- B2. From MeteoSwiss Basel-Binningen station (station identifier BAS):
 - CDD22 (base temperature 22.0 degrees C) for the 1991-2020 climate norm.
 - CDD22 for years 2019, 2022, 2023, and 2024 individually.
 - CDD24 for the same years for comparison.

- Peak July representative hourly temperature and global radiation profile or the dataset identifier from which this can be retrieved.
- Frequency of days per year exceeding 22.0 and 30.0 degrees C, averaged over 1991–2020 and for recent exceptional years.
- Summer night minimum temperatures: frequency of tropical nights ($T_{\min}$ above 20 degrees C) per year and trend data.

B3. From ASHRAE HVAC Systems and Equipment Handbook: state the edition year, chapter number, and table number for equipment service life data. Provide median lifetimes for: hermetic compressors, electronic control boards, condenser and evaporator coil assemblies, fan motors, expansion valves.

B4. Retrieve AIVC Airbase Paper 13390 on buoyancy-driven flow through a stairwell. Confirm it is accessible at aivc.org. State the volumetric airflow formula for a vertical shaft natural convection opening and the discharge coefficient C_d recommended for residential staircases.

SECTION C - APPLIANCE-SPECIFIC DATA

C1. Miele K 37682 iDF:

- Confirm exact product type: integrated refrigerator with small freezer compartment, not a standalone freezer.
- EU energy label rated annual consumption in kWh per year at standard 25 degrees C ambient test condition per IEC 62552 or EN ISO 15502.
- Energy efficiency class under the 2021 EU label scale.
- Net refrigerator volume in litres; net freezer compartment volume if any.
- Data source and document reference URL.

C2. Liebherr GNP 3166:

- Confirm exact product type: standalone upright freezer (Gefriergerät), not a fridge-freezer combination.
- EU energy label rated annual consumption at 25 degrees C ambient.
- Energy efficiency class.
- Data source and document reference URL.

C3. From IEC 62552-3:2015 or peer-reviewed study: the measured sensitivity of household refrigerator energy consumption to ambient temperature rise above 25 degrees C, expressed as percent increase per Kelvin. Provide the specific value and a full citation with DOI. Confirm the DOI resolves.

SECTION D - LIFECYCLE ASSESSMENT DATA

D1. Search the ECO Platform database at ecosmdp.org and the IBU EPD database at ibu-epd.com for any EPD registered for a Daikin wall-mounted residential split-type air conditioning product in the 3.5 to 6.0 kW capacity class. Report the search result in full: registration number, product name, product class, capacity range, and publishing organisation if found; or an explicit statement that no such product-specific EPD was found. If only a proxy from a different product class is available, state this explicitly and quantify the product class differences.

D2. Search for at minimum two peer-reviewed ISO 14040/14044-compliant LCA studies for residential split-type air conditioners in the 3.5 to 5.5 kW capacity class published after 2015. For each study provide: full citation including author names, journal name, volume, pages, year, and DOI. Confirm each DOI resolves. Extract: cradle-to-gate embodied carbon (Modules A1–A3) in kg CO₂e; total primary grey energy in MJ.

D3. From Swiss FOEN/BAFU or SENS eRecycling published statistics: the actual end-of-life refrigerant recovery rate achieved in Switzerland for residential refrigerating and air conditioning appliances. Provide the document title, year, and specific table or page number.

D4. From EU Regulation 517/2014 or 2024/573: the default annual leak rate for stationary split air conditioning systems containing less than 3 kg of refrigerant, as used for emissions inventory calculations. State the specific annex, article, or table.

D5. From IPCC 2006 Guidelines for National Greenhouse Gas Inventories: the default annual leak rate for residential split air conditioning

systems used as the high-end or worst-case assumption. State the chapter and table number.

SECTION E - ENERGY AND GRID DATA

- E1. From SFOE published statistics: Swiss domestic electricity production mix carbon intensity in g CO₂e per kWh and Swiss electricity consumption mix carbon intensity in g CO₂e per kWh. State the document title, year, and table or figure reference for each.
- E2. From ENTSO-E Transparency Platform or peer-reviewed study: the estimated summer daytime marginal carbon emission factor for Switzerland in g CO₂e per kWh. State the dataset name, time period, and methodology. Assess whether Swiss summer daytime marginal generation is set by gas, hydro, nuclear, or cross-border imports.
- E3. From SFOE or SIA 380/1: the primary energy conversion factor for Swiss grid electricity. State the document reference and year.

SECTION F - DAIKIN TECHNICAL SPECIFICATIONS

- F1. From Daikin technical databook for the FTXA/RXA series: cite the document reference number as it appears on the document cover page. Retrieve for both FTXA42 and FTXA50:
- Rated cooling capacity at Condition A (35 degrees C outdoor).
 - Energy Efficiency Ratio (EER) at Condition A.
 - EER at Condition B (30 degrees C outdoor).
 - Indoor unit high-speed airflow rate in m³ per minute or m³ per second.
 - Sensible Heat Ratio (SHR) at standard test conditions if stated.
 - Rated evaporating temperature at Condition A if stated.
 - Rated condensing temperature at Condition A if stated.
 - WiFi-connected standby power draw of indoor unit in watts.
 - Outdoor unit stator winding trickle heating power in watts if stated.
 - SEER rating.
- F2. From Daikin RXA42 or RXA50 installation or service manual: cite the exact document reference number. Retrieve:
- Required minimum pre-season power-on duration before first compressor operation after electrical disconnection.
 - Confirmation of whether physical crankcase heater band is fitted or stator winding heating is used instead.
 - If stator winding heating: the rated power in watts.

SECTION G - BEHAVIOURAL AND HEALTH DATA

- G1. From Swiss Tropical and Public Health Institute (Swiss TPH), WHO, or peer-reviewed Swiss epidemiological literature: indoor temperature thresholds above which heat-related health risks become clinically significant for healthy adults and for vulnerable populations. Provide full citations with DOIs.
- G2. From Swiss federal or cantonal water statistics: average shower water consumption in litres per shower event. State the document title, organisation, and year.
- G3. From SFOE or EU energy label database: typical energy consumption per cycle for a washing machine at 60 degrees C cotton programme and for a condensation type tumble dryer. State values in kWh per cycle and source.
- G4. From peer-reviewed literature: lifecycle embodied carbon of a representative smartphone and a representative laptop computer in kg CO₂e per device. Provide full citations with DOIs. Confirm DOIs resolve.
- G5. Search peer-reviewed literature or published lifecycle assessment databases for embodied carbon of a standard cotton bedding set in kg CO₂e. If no acceptable source is found, state [NOT FOUND] explicitly.
- G6. From peer-reviewed literature: estimated carbon footprint per acute emergency hospitalisation event in Switzerland or comparable European healthcare system in kg CO₂e. Provide the citation with DOI.

SECTION H - PARQUET FLOORING EMBODIED CARBON

H1. From a peer-reviewed LCA study or verified EPD for hardwood parquet flooring: the cradle-to-grave or cradle-to-gate embodied carbon in kg CO₂e per m² of installed parquet. Provide citation with DOI or EPD registration number. If no acceptable source is found, state [NOT FOUND].

SECTION I - R-32 SOLUBILITY IN POE LUBRICANT

I1. From peer-reviewed refrigeration engineering literature: data on the solubility of R-32 in polyolester (POE) compressor lubricating oil as a function of temperature at 0 to 15 degrees C. Provide the source with full citation. Acceptable sources: International Journal of Refrigeration, ASHRAE Transactions.

OUTPUT FORMAT - MANDATORY

Produce the output as a structured data repository. Under each data point:

Line 1: DATA POINT NAME in capitals

Line 2: VALUE: the numerical value or qualitative finding

Line 3: SOURCE: full citation with author, title, publication, year, DOI or URL, document reference number, clause or table number as applicable

Line 4: VERIFICATION STATUS: VERIFIED / UNVERIFIED DOI / APPROXIMATED / NOT FOUND

Line 5: UNCERTAINTY: stated as percentage range or qualitative description

Line 6: NOTES: caveats, conflicts with other sources, flags for modelling

After all sections append a GAPS AND FLAGS summary listing every data point where VERIFICATION STATUS is anything other than VERIFIED.

Do not attempt to calculate, model, synthesise, or write any section of the sustainability report. Record and report only.

A.2. Prompt 2: Claude Sonnet — Full Modelling and Report Generation Prompt

Prompt 2: Claude Sonnet — complete verbatim text

You are conducting a rigorous quantitative sustainability modelling exercise. You will receive two inputs: (1) a structured data repository produced by a prior deep research phase containing all externally sourced data with citations and verification status, and (2) a set of confirmed owner-measured parameters. Your task is to use these inputs to produce a comprehensive sustainability assessment report.

CRITICAL OPERATING PRINCIPLE

You must not conduct any new research. You must not retrieve any information from your training memory to fill gaps in the provided data. Every quantitative value you use must come either from the data repository or from the confirmed parameters block below, or must be derived by arithmetic or physics from those sources. If a required input is missing from both sources, state explicitly: [MISSING INPUT - calculation cannot be completed without X, which was not provided in the data repository.]

CONFIRMED FIXED PARAMETERS - INVIOABLE THROUGHOUT THE REPORT

Indoor thermostat setpoint: 22.0 degrees C [MEASURED].

Peak outdoor shade temperature: 39.0 degrees C [MEASURED].

Peak temperature differential roof to indoor: $45.0 - 22.0 = 23.0$ K [CALC].

Peak temperature differential wall to indoor: $39.0 - 22.0 = 17.0$ K [CALC].

Peak daily electrical consumption upper bound: 7.80 kWh per day [MEASURED].

Grid draw on active cooling days: zero [MEASURED].

Water heating: heat pump water heater, COP 3.0, owner-confirmed [MEASURED].

Use 0.071 kg CO₂e per shower (1.66 kWh / COP 3.0 x 0.128 kg/kWh).

Do NOT use gas boiler values for shower calculations.

Battery storage: Home battery system covers household overnight electricity

demand during May–September connected months [MEASURED].
 Nighttime standby draw (26.5 W) is supplied from stored PV electricity.
 Grid carbon for nighttime standby = zero.
 Module B6 = 0 kg CO₂e per year. 15-year B6 total = 0 kg CO₂e.
 Do NOT assign any grid carbon to nighttime standby.
 State explicitly in the standby section: "Module B6 = 0 because
 (a) October–April disconnection and (b) home battery covers May–Sep
 nighttime standby. Grid carbon: zero."
 Passive cooling strategy: Owner employs night ventilation, morning pre-cooling,
 and daytime window and blind closure per Basel city guidance [MEASURED].
 Active AC is used only as a residual measure when passive strategies are
 insufficient. The confirmed 7.80 kWh/day peak consumption already reflects
 this combined passive-plus-active approach. Include a dedicated subsection
 describing this strategy and its significance for interpreting the results.
 System: Daikin Stylish FTXA42 or FTXA50 with RXA-A or RXA-B outdoor unit.
 Refrigerant: R-32, factory charge 1.30 kg [MEASURED].
 Safety concentration limit: 0.061 kg per cubic meter.
 OG floor area: 60.8 m² [MEASURED]. OG volume: 145.9 m³ [MEASURED].
 OG ceiling: 2.40 m [MEASURED].
 EG floor area: 60.1 m² [MEASURED]. EG volume: 151.9 m³ [MEASURED].
 EG ceiling: 2.53 m [MEASURED]. Use 2.53 m as H in all staircase formulas.
 UG floor area: approximately 48.0 m². UG volume: approximately 108.4 m³.
 Total conditioned volume: approximately 297.8 m³.
 OG parquet floor area: approximately 49.0 m² [MEASURED].
 Staircase: open vertical shaft; not a horizontal vent.
 Building: 1999/2000, SIA 380/1:1988, owner-described as relatively well
 insulated.
 Operational pattern: completely powered off October through April;
 energised May through September only [MEASURED].
 Miele K 37682 iDF: EG kitchen location [MEASURED].
 Rated consumption 230–234 kWh/year at 25 degrees C [FROM REPOSITORY].
 Integrated refrigerator with small freezer compartment.
 Liebherr GNP 3166: UG basement location [MEASURED].
 Standalone upright freezer.
 Basement 16–20 degrees C year-round from ground thermal mass [MEASURED].
 AC system has zero influence on basement temperature.
 Liebherr MUST be excluded from all avoided carbon calculations.
 Swiss grid consumption mix carbon factor: 128 g CO₂e per kWh
 [FROM REPOSITORY: E1].
 Swiss gas emission factor: 0.201 kg CO₂e per kWh thermal
 [FROM REPOSITORY: E1]. Note: used only for gas-heated items; the
 confirmed heat pump water heater uses the electricity factor not gas.
 Daily PV surplus exported to grid: unknown; name this S (kWh per day).
 Treat S as a sensitivity parameter. Range for analysis: 5 to 20 kWh/day.
 Never assign S a specific numerical value except in clearly labelled
 illustrative examples.

SOURCE TAGGING RULES

Every quantitative claim must carry one of four inline tags:
 [MEASURED] – owner-confirmed metered data from confirmed parameters above.
 [CALCULATED] – derived by arithmetic or physics from stated inputs; show.
 [FROM REPOSITORY: code] – taken directly from the data repository.
 [APPROXIMATED: pct%] – engineering estimate; state percentage uncertainty.
 [MISSING INPUT] – required value absent from parameters and repository.

CONSISTENCY RULES – ENFORCED THROUGHOUT

Every number appearing in more than one section must be identical in every
 appearance. Before writing each section check against all prior sections.
 Before writing the executive summary verify every number against its section.
 Correct any discrepancy immediately and note it.

Define once and use identically everywhere:

- Corrected real-world COP range from Section 1b.
- Annual compressor run hours from Section 1b.
- Central annual active cooling energy from Section 3b.
- Nighttime standby annual energy and carbon from Section 3b (both zero).
- Upfront embodied carbon A1–A5 from Section 3a.
- Refrigerant leakage best-practice 15-year AR5 and AR6 values from Sec 2b.
- EOL carbon C1–C4 from Section 4c.
- Behavioural carbon total from Section 8e (using heat pump WH values).

- Module D annual coefficient from Section 5a.
- Net carbon formula: $C_{net} = +2.9 - 9.0 \cdot S$ kg CO₂e/yr (before S known).
- Zero-carbon threshold: $S^* = 0.32$ kWh/day.

RULE 1 - No invented numbers. S is never given a specific value except in clearly labelled illustrative examples.

RULE 2 - Mandatory cross-check. Construct the consistency appendix before finalising. Every value in multiple sections must be verified identical.

Three mandatory cross-calculations:

- (1) Section 4c AR6 15yr total + Section 5a B6 15yr total = Section 5a total.
Note B6 = 0, so Section 4c total should equal Section 5a total.
 - (2) Section 5b Daikin row total = Section 5a Systemic Total excluding D.
 - (3) Executive summary behavioural carbon = Section 8e central total exactly.
- Report any discrepancy exceeding 0.5 percent as requiring correction.

RULE 3 - Source tag every quantitative claim per the tagging rules above.

RULE 9 - COP physical validity. Every derived COP must not exceed the Level 2 Carnot COP from Section 1a. Flag PHYSICALLY INCONSISTENT and resolve by dynamic compressor power using Daikin EN 14825 EER data. Show arithmetic.

RULE 10 - Standby mandatory quantification. Both October–April and May–September nighttime standby = zero grid carbon. State this explicitly. October–April: zero due to physical disconnection [MEASURED].

May–September nighttime: zero because home battery covers overnight draw [MEASURED]. Module B6 = 0 kg CO₂e per year and 0 kg CO₂e over 15 years.

RULE 11 - R-32 production carbon consistency. State the virgin R-32 production intensity once in Section 2 with citation. Use identically everywhere. State whether included in or additional to EPD value.

RULE 12 - Module D parametric. Module D = $-S \times t_{export} \times EF_{marginal} \times LT$. Define t_{export} and $EF_{marginal}$ once in Section 5a. Use these identical values in Sections 6b and 9. Module D table row must contain the formula only, never a single number.

RULE 13 - Scorecard scale: Excellent, Good, Adequate, Poor, Critical only.

RULE 14 - Blocked domains: reddit.com, r32reasons.com, hvacprosales.com, bluewin.ch, wetter.li, Scribd, Academia.edu, mckinsey.com, eia-international.org, openco2.net, tsclair.com, total-environmental.co.uk, and any HVAC installer website not owned by the original equipment manufacturer.

RULE 15 - EPD verification. Before citing any EPD confirm its registration number corresponds to a wall-mounted residential split-type AC product via ecosmdp.org or ibu-epd.com. If not found state [NOT FOUND] explicitly.

RULE 16 - Energy payback basis. State whether final energy or primary energy basis is used. Apply consistently in both numerator and denominator.

RULE 17 - Citation self-audit. Before finalising references construct Appendix B citation audit table: reference number, author/organisation, domain, source category, Rule compliance Pass/Fail, replacement action. Remove every failing reference. Minimum 25 clean references.

SECTION 1 - THERMODYNAMIC EFFICIENCY AND REAL-WORLD PERFORMANCE

1a. Calculate Level 1 Carnot COP using $T_{cold} = 22.0$ degrees C = 295.15 K and $T_{hot} = 39.0$ degrees C = 312.15 K. Show arithmetic fully.

Derive Level 2 Carnot COP from first principles using FTXA airflow, SHR = 0.75, approach temps. Note 22.0 degrees C setpoint effect. Apply condensing lift 11–16 K above 39.0 degrees C. Calculate Level 2 Carnot COP at both T_{cond} bounds. State Level 2 is the correct reference. List all irreversibility sources with efficiency penalties.

1b. Building thermal load and real-world COP.
Use $T_{indoor} = 22.0$ degrees C throughout all calculations.
Paths 1–4 as specified. Diurnal average load from MeteoSwiss BAS data

(Rule 8 - non-circular). COP physical validity per Rule 9.
Derive annual compressor run hours T_{run} for use in Section 4.

1c. Staircase convection using AIVC 13390 formula. $H = 2.53 \text{ m}$ [MEASURED].
Note 22.0 degrees C setpoint enhances staircase flow vs 24.0 degrees C.

SECTION 1 ADDITIONAL - PASSIVE COOLING STRATEGY AND OPERATIONAL CONTEXT

Include a dedicated subsection titled "Passive Cooling Strategy and Operational Context" describing:

- The night ventilation and morning pre-cooling practice.
- Daytime window and blind closure.
- That active AC is residual-only, per Basel city guidance.
- That the 7.80 kWh/day peak already reflects combined approach.
- That MeteoSwiss tropical night data shows passive ventilation is increasingly insufficient on the most extreme days.

This subsection does not affect any numerical calculations.

SECTION 2 - REFRIGERANT R-32 ENVIRONMENTAL PROFILE

As previously specified. R-410A AR5 GWP = 2088 (corrected).

SECTION 3 - GREY ENERGY AND EMBODIED CARBON

3b. Annual operational energy includes standby discussion:

State that Module B6 = 0 kg CO₂e per year and 0 kg CO₂e over 15 years for two independent reasons: (a) October-April physical disconnection and (b) home battery system covers May-September nighttime standby draw. Grid carbon for all standby periods = zero. Do not calculate any standby carbon. Do not include any standby term in Module B6.

3c. Carbon payback: numerator = A1-A5 upfront carbon only.

Denominator = annual avoided operational carbon = active cooling energy x grid factor. Standby cancels in comparison (both zero for this system).

SECTION 4 - OPERATIONAL LIFETIME

As previously specified. Note that standby B6 = 0.

SECTION 5 - NET LIFECYCLE CARBON BALANCE

5a. EN 15978 table. B6 row must show 0 kg CO₂e (battery-covered) for all lifetimes and both AR5 and AR6. Note: "Home battery covers nighttime standby; grid carbon = zero." Total excluding Module D = 497 kg CO₂e (15yr, AR6). Annualised = 33.1 kg CO₂e/yr.

Module D: define coefficient = $t_{\text{export}} \times EF_{\text{marginal}}$ once. Formula only in the table row; never a single number.

SECTION 6 - URBAN HEAT ISLAND AND GRID INTEGRATION

As previously specified. Include the dedicated heat rejection physics box.

SECTION 7 - SUSTAINABILITY VERDICT

As previously specified. R5 recommendation: winter isolation already implemented; battery covers summer nighttime standby. R6: passive cooling strategy is already in place and consistent with Basel city guidance. Note in the recommendations that the AC is used as a residual measure only.

SECTION 8 - BEHAVIOURAL CARBON EXTERNALITIES

Use heat pump WH (COP 3.0); shower carbon = 0.071 kg CO₂e per shower.

Liebherr explicitly excluded with explanation.

Total behavioural avoided carbon central = 30.2 kg CO₂e/yr.

SECTION 9 - INTEGRATED SUSTAINABILITY VERDICT

Net formula: $C_{\text{net}} = 33.1 - 30.2 - 9.0 \times S = +2.9 - 9.0 \times S$ kg CO₂e/yr.

Zero-carbon threshold: $S^* = 2.9 / 9.0 = 0.32 \text{ kWh/day}$.

State that this threshold is met within approximately 4 minutes of peak summer solar generation.

State that 2.9 kg CO₂e/yr is equivalent to driving a small petrol car approximately 21 km (Basel city centre to Pratteln).

Note that with the battery system confirmed, the system is net-negative on essentially every connected day with any solar generation.

SECTION 10 - LIMITATIONS DISCLOSURE

Standby bullet: Module B6 is zero because of both physical disconnection and battery coverage. The +/-50% standby power uncertainty is irrelevant to the carbon calculation.

SECTION 11 - MANDATORY CONSISTENCY APPENDIX

All values verified. B6 = 0 throughout. Net formula verified as $C_{net} = +2.9 - 9.0 \cdot S$.

APPENDIX B - CITATION SELF-AUDIT TABLE (RULE 17)

Every reference audited. Minimum 25 clean references.

EXECUTIVE SUMMARY - WRITE THIS ABSOLUTELY LAST

Maximum 400 words. State: 22.0 degrees C setpoint; heat pump water heater reduces shower carbon to 10 kg CO₂e/yr; home battery eliminates standby carbon (B6 = 0); winter disconnection; passive cooling strategy per Basel city guidance with AC as residual-only measure; net formula $+2.9 - 9.0 \cdot S$; $S^* = 0.32$ kWh/day; 2.9 kg CO₂e/yr = 21 km drive; net-negative on virtually every connected day. Do not use the word sustainable. Do not claim zero carbon. State the full lifecycle footprint of 497 kg CO₂e.

B. Air Conditioning, Heat Rejection, and Urban Atmospheric Warming

B.1. Introduction

A common objection to residential air conditioning is that AC units “heat up the neighbourhood.” This appendix addresses that claim rigorously, using first-principles thermodynamics, the Basel BUBBLE field experiment [20], and peer-reviewed urban climatology literature [21, 22, 23, 24].

B.2. First Law of Thermodynamics

A heat pump does not generate heat; it moves thermal energy from a cold reservoir to a hot reservoir. The energy balance at the outdoor condenser unit is governed by the First Law of Thermodynamics:

$$\dot{Q}_{\text{rejected}} = \dot{Q}_{\text{absorbed, indoor}} + W_{\text{electrical}} \quad (1)$$

$\dot{Q}_{\text{absorbed, indoor}}$ — the heat extracted from the building interior — was already present in the local thermal environment before the AC cycle began. Relocating it outdoors does not create new heat. The **only net addition of thermal energy to the outdoor environment** is the electrical work:

$$W_{\text{electrical}} = \dot{Q}_{\text{cooling}} / \text{COP} = 3.00 / 3.85 = 0.779 \text{ kW}$$

In this installation, even this 0.779 kW is supplied by photovoltaic conversion of solar radiation — energy that would otherwise have been absorbed as heat by the roof surface regardless of whether the AC operates.

B.3. Spatial Scale: Basel Urban Heat Island Contribution

The Basel BUBBLE experiment [20] measured anthropogenic heat flux in residential areas at **5–25 W/m²** (mean ≈ 15 W/m² during summer daytime). Peak heat rejection at the building footprint:

$$q_{\text{footprint}} = (3.00 + 0.779) \text{ kW}/60.8 \text{ m}^2 = 62 \text{ W/m}^2$$

Seasonal average over the May–September connected period:

$$q_{\text{seasonal}} = \frac{378 \text{ kWh/yr}}{60.8 \text{ m}^2 \times 153 \text{ days} \times 24 \text{ h/day}} = 1.69 \text{ W/m}^2$$

At typical Basel suburban residential density (20 buildings/hectare, 20% footprint ratio [24, 22]), the neighbourhood-scale seasonal average is $1.69 \times 0.20 = 0.34 \text{ W/m}^2$, representing $\approx 2\%$ of the BUBBLE residential background. Allen et al. [23] confirmed that at current European residential AC penetration rates, direct heat from AC is secondary to urban morphology (impervious surfaces, reduced vegetation, canyon geometry) as the dominant urban heat island driver.

title

Does this unit heat the neighbourhood? The seasonal neighbourhood-scale contribution of 0.34 W/m^2 is approximately 2% of the BUBBLE residential background flux of 15 W/m^2 . Real but negligible at city scale. The unit is undetectable in any urban climate measurement.

Does daytime-only operation matter for UHI? Yes, critically. The nocturnal stable boundary layer ($< 100 \text{ m}$ depth) traps anthropogenic heat near the surface where it affects sleeping residents. The daytime convective boundary layer ($> 1,000 \text{ m}$) rapidly disperses heat vertically. Because this system operates exclusively on solar power, it rejects **zero heat at night** — the optimal operating pattern from an urban climate perspective. Grid-powered units that run through the night contribute meaningfully to nocturnal UHI; this solar-coupled unit does not [25].

Does heat rejection contribute to global warming? No, in any physically meaningful sense. Moving thermal energy from building interior to the outdoor atmosphere does not change the total thermal energy of the climate system [26]. The fraction of atmospheric thermal energy contributed by this unit annually is approximately 10^{-15} — physically indistinguishable from zero.

The *only* climate-relevant mechanism is greenhouse gas emissions from electricity generation. For this installation: active cooling GHG = $0 \text{ kg CO}_2\text{e/yr}$ (100% PV); nighttime standby GHG = $0 \text{ kg CO}_2\text{e/yr}$ (battery-covered); total operational GHG = $0 \text{ kg CO}_2\text{e/yr}$.

Key references: Rotach et al. [20]; Oke et al. [21]; Sailor [22]; Allen et al. [23]; IPCC AR6 [26].

References

- [1] IPCC. *Sixth Assessment Report, WGI, Chapter 7, Table 7.SM.7*. Assessment Report. Intergovernmental Panel on Climate Change, 2021.
- [2] D. Europe. *Technical Databook EEDEN18, RXA-B series*. daikintech.co.uk. 2019.
- [3] MeteoSwiss. *Basel-Binningen Station BAS: Climate Normals 1991–2020*. Accessed 2026. 2026.
- [4] S. S. of Engineers and Architects. *SIA 380/1:1988 Thermal Energy in Building Construction*. SIA, 1988.
- [5] AIVC. *Buoyancy-Driven Flow Through a Stairwell*. Tech. rep. Airbase Paper 13390. Air Infiltration and Ventilation Centre, 2000.
- [6] MeteoSwiss. *Scientific Report No. 105: Urban Heat Island Future Climates*. MeteoSwiss, 2020.
- [7] IPCC. *Fifth Assessment Report, WGI, Chapter 8, Table 8.SM.16*. Assessment Report. Intergovernmental Panel on Climate Change, 2013.
- [8] S. eRecycling. *Fachbericht 2024*. SENS eRecycling, 2024.
- [9] E. Union. *EU F-Gas Regulation 2024/573, Article 2; Annex IV*. 2024.
- [10] S. S. Association. *SN EN 378-1:2016 Refrigerating Systems and Heat Pumps*. Swiss Standards Association, 2016.
- [11] F. Sosa et al. “Lifecycle Carbon of R-32 Production”. In: *Sustainable Materials and Technologies* 39 (2024), e00811. DOI: [10.1016/j.susmat.2023.e00811](https://doi.org/10.1016/j.susmat.2023.e00811).
- [12] OEKOBAUDAT. *Process Dataset: Difluoromethane R-32*. ID: c34de763-3a6c-437d-9b3a-566926f8abc4. 2024.
- [13] T. Kaneko et al. “R-32 Solubility in Polyolester Lubricant”. In: *International Journal of Refrigeration* 111 (2019).
- [14] IEC. *IEC 62552-3:2015 Household Refrigerating Appliances*. International Electrotechnical Commission, 2015.
- [15] Miele. *Product Datasheet: K 37682 iDF Integrated Refrigerator*. EU energy label: 230–234 kWh/year. 2023.
- [16] S. F. O. of Energy. *Schweizerische Gesamtenergiestatistik*. SFOE, 2024.
- [17] ENTSO-E. *Transparency Platform: Swiss Summer Marginal Emission Factor*. Accessed 2026. 2025.
- [18] ASHRAE. *HVAC Systems and Equipment Handbook, Chapter 37, Table 4*. 2020th ed. 2020.
- [19] ISO. *ISO 281:2007 Rolling Bearings: Dynamic Load Ratings*. ISO, 2007.
- [20] M. Rotach, R. Vogt, C. Bernhofer, et al. “BUBBLE: an Urban Boundary Layer Meteorology Project”. In: *Theoretical and Applied Climatology* 81 (2005), pp. 231–261. DOI: [10.1007/s00704-004-0117-9](https://doi.org/10.1007/s00704-004-0117-9).
- [21] T. Oke, G. Mills, A. Christen, and J. Voogt. *Urban Climates*. Cambridge University Press, 2017. DOI: [10.1017/9781139016589](https://doi.org/10.1017/9781139016589).
- [22] D. Sailor. “A Review of Methods for Estimating Anthropogenic Heat”. In: *International Journal of Climatology* 31 (2011), pp. 189–199. DOI: [10.1002/joc.2106](https://doi.org/10.1002/joc.2106).

- [23] L. Allen, F. Lindberg, and C. Grimmond. “Global to City Scale Urban Anthropogenic Heat Flux”. In: *International Journal of Climatology* 31 (2011), pp. 1990–2005. doi: [10.1002/joc.2210](https://doi.org/10.1002/joc.2210).
- [24] C. Grimmond. “The Suburban Energy Balance”. In: *International Journal of Climatology* 12 (1992), pp. 481–497.
- [25] S. Hassid, M. Santamouris, et al. “The Effect of the Athens Heat Island on AC Load”. In: *Energy and Buildings* 32 (2000), pp. 131–141. doi: [10.1016/S0378-7788\(99\)00045-6](https://doi.org/10.1016/S0378-7788(99)00045-6).
- [26] IPCC. *AR6 WGI Chapter 7: The Earth’s Energy Budget*. Intergovernmental Panel on Climate Change, 2021.